

STATS 7022 - Data Science

Week 2: Regression and Classification

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Regression Recap

MLR

Consider data

$$(y_1, x_{11}, x_{12}, \dots, x_{1p}), (y_2, x_{21}, x_{22}, \dots, x_{2p}), \dots, (y_n, x_{n1}, x_{n2}, \dots, x_{np}).$$

The multiple regression model postulates

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i,$$

where ϵ_i are i.i.d. with

$$E(\epsilon_i) = 0 \text{ and } \text{var}(\epsilon_i) = \sigma^2.$$

MLR Matrix Form

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & & & & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}, \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

The multiple regression model can then be formulated as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

with

$$E(\boldsymbol{\epsilon}) = \mathbf{0} \text{ and } \text{var}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}_{n \times n}.$$

Least Squares Estimates

The least squares estimate $\hat{\beta}$ is the vector that minimises the sum of squares

$$Q(\beta) = \|\mathbf{y} - X\beta\|^2.$$

The variance σ^2 is estimated by

$$s_e^2 = \frac{1}{n - p - 1} \|\mathbf{y} - X\hat{\beta}\|^2.$$

where $p + 1$ is the number of columns of X .

Theorem

If the columns of X are linearly independent then the least squares estimates are given uniquely by

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.$$

Some Important Questions

Q1: Is There a Relationship Between Response and Predictors?

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0,$$

$$H_a : \text{at least one } \beta_j \text{ is non-zero.}$$

$$F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)},$$

where

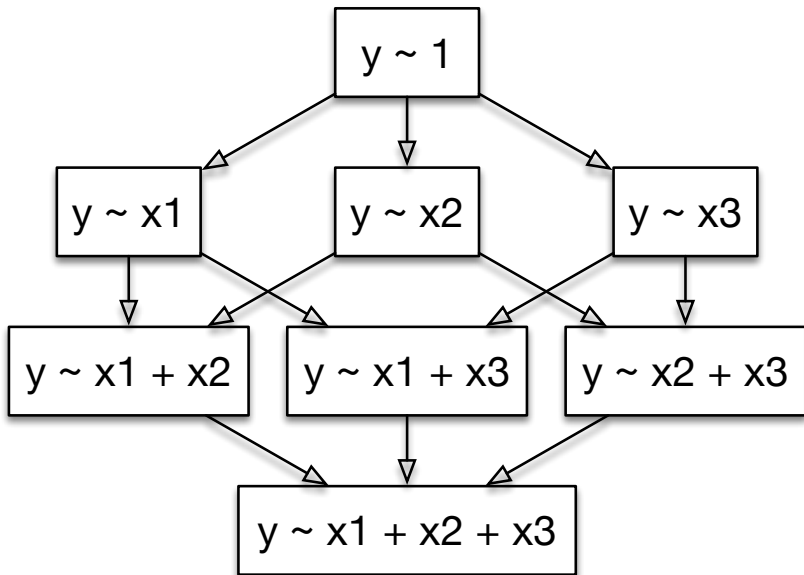
$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2 \text{ and } RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Q2: Deciding on Important Variables

Quality of model: AIC, BIC, AICc, adjusted R^2 .

Approaches:

1. Forward selection: Start with null and add.
2. Backwards selection: Start with full and remove.
3. Mixed selection: start with null and add, but can also remove.



Q3: Model Fit

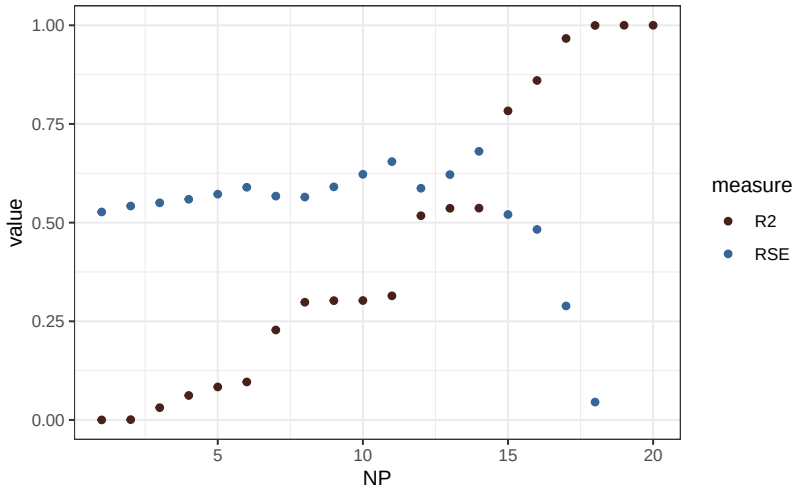
Classical measures are

$$R^2 = \frac{TSS - RSS}{TSS}$$

and

$$RMSE = \sqrt{\frac{RSS}{n - p - 1}}.$$

Warning



Warning

```
set.seed(2020)
Y <- rep(0:1, 10)
X <- matrix(rnorm(20*20), ncol = 20)

1:20 %>%
  map_df(~broom::glance(lm(Y ~ X[, 1:.x])), .id = "i") %>%
  mutate(i = as.numeric(i)) %>%
  select(R2 = r.squared, RSE = sigma, NP = i) %>%
  gather(key = "measure", value = value, -NP) %>%
  ggplot(aes(NP, value, col = measure)) + geom_point() +
  scale_color_paletteer_d("dutchmasters::milkmaid")
```

Q4: Predictions

Consider a new point $(Y_0, \mathbf{x}_0)$.

The $100(1 - \alpha)\%$ confidence interval for $E[Y_0]$ is

$$\mathbf{x}_0^T \hat{\beta} \pm t_{n-p-1}(\alpha/2) s_e \sqrt{\mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0}.$$

The $100(1 - \alpha)\%$ prediction interval for Y_0 is

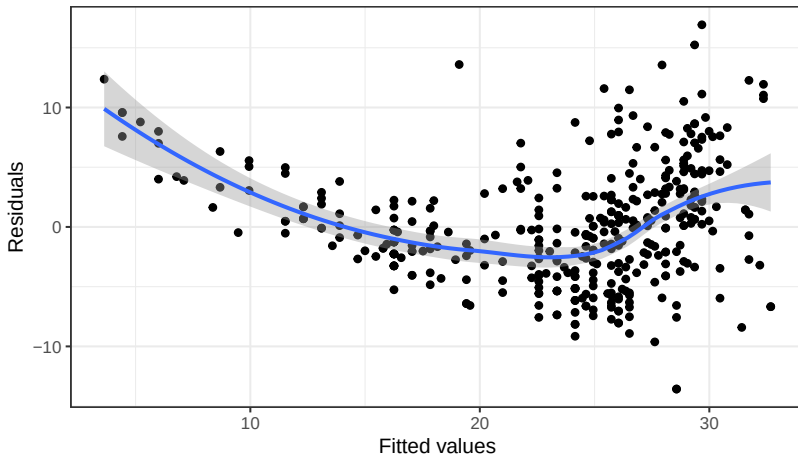
$$\mathbf{x}_0^T \hat{\beta} \pm t_{n-p-1}(\alpha/2) s_e \sqrt{1 + \mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0}.$$

Potential Problems

Non-linearity of Data

- ▶ Residuals versus fitted plot.
- ▶ MLR: residual versus each predictor.

Residual versus fitted



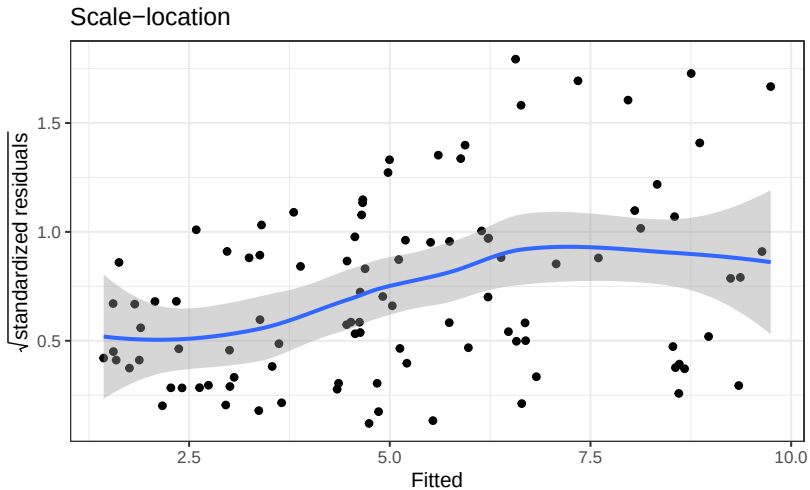
Correlation of Error Terms

We assume that the error terms are uncorrelated. If they are correlated, then the estimated standard errors of estimators tend to underestimate the true standard errors. This leads to confidence intervals and prediction intervals that are narrower than they should be.

Common Causes of Correlation

- ▶ Clustering
- ▶ Temporal
- ▶ Spatial

Non-constant Variance of Error Terms



Transformation

Try transforming the response variable, *i.e.*, $\log(Y)$, $\sqrt{Y}$.

Weighted Least Squares

We use a $n \times n$ weighting matrix W , with case weights w_i on the diagonal and zeros elsewhere.

The multiple regression model can then be formulated as

$$W^{1/2} \mathbf{Y} = W^{1/2} X \boldsymbol{\beta} + W^{1/2} \mathbf{e}$$

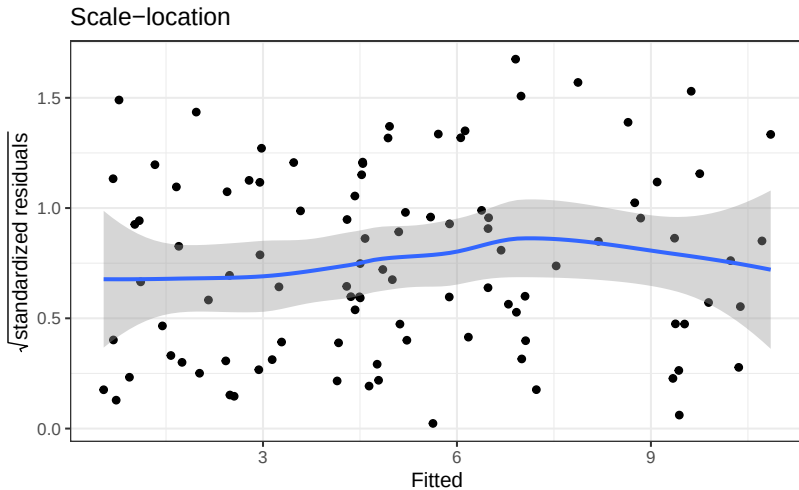
and

$$\hat{\boldsymbol{\beta}} = ((X^*)^T X^*)^{-1} X^{*T} \mathbf{y}^*,$$

where

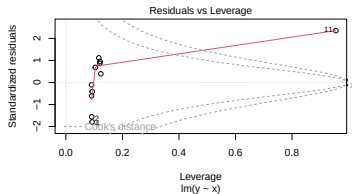
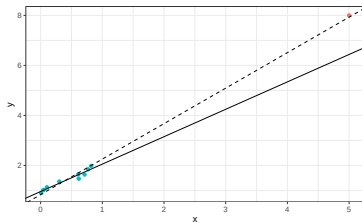
$$X^* = W^{1/2} X \text{ and } \mathbf{y}^* = W^{1/2} \mathbf{y}.$$

Weighted Least Squares

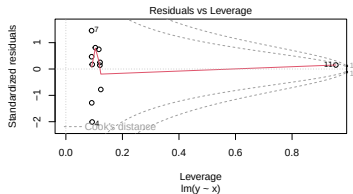
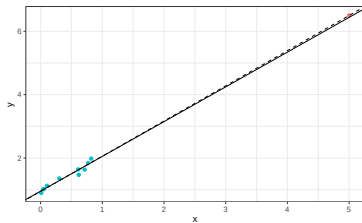


Influential Points

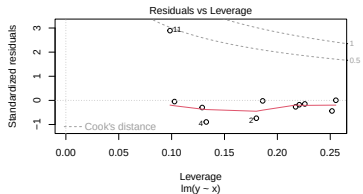
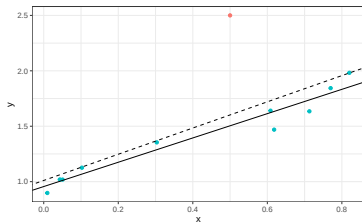
Large Residual, High Leverage



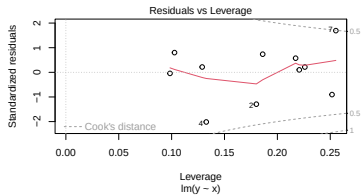
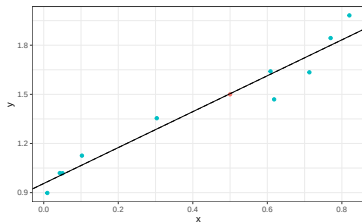
Small Residual, High Leverage



Large Residual, Low Leverage



Small Residual, Low Leverage

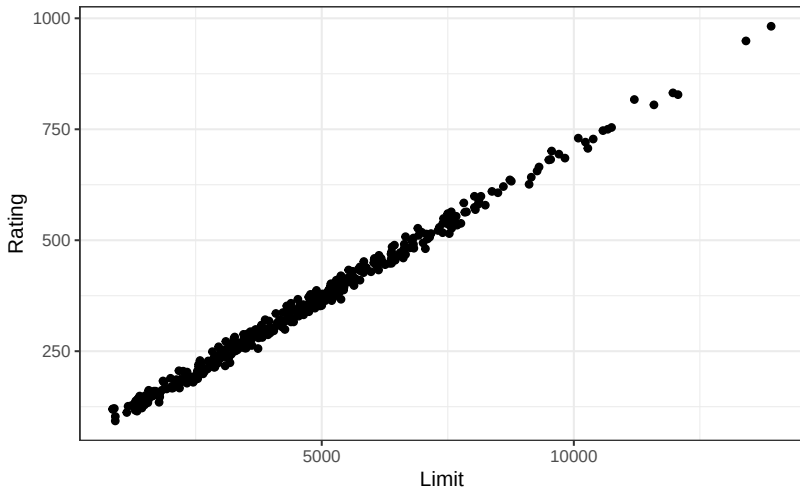


Collinearity

Collinearity refers to the case where two predictors are closely related.

Collinearity can cause the standard errors of the coefficient estimates to grow.

Example



Example

term	estimate	std.error	statistic	p.value
Model 1				
(Intercept)	-173.4109014	43.8283870	-3.9565887	0.0000901
Age	-2.2914855	0.6724845	-3.4074918	0.0007226
Limit	0.1733650	0.0050257	34.4959445	0.0000000
Model 2				
(Intercept)	-377.5367954	45.2541762	-8.3425846	0.0000000
Rating	2.2016722	0.9522939	2.3119672	0.0212905
Limit	0.0245144	0.0638346	0.3840298	0.7011619

Variance Inflation Factor (VIF)

The VIF is the ratio of the variance of $\hat{\beta}_j$ when fitting the full model divided by the variance of $\hat{\beta}_j$ if fit on its own.

The smallest possible value of the VIF is 1.

A VIF greater than 5 indicates problems.

Example Again

##	Age	Limit	Rating
##	1.011385	160.592880	160.668301

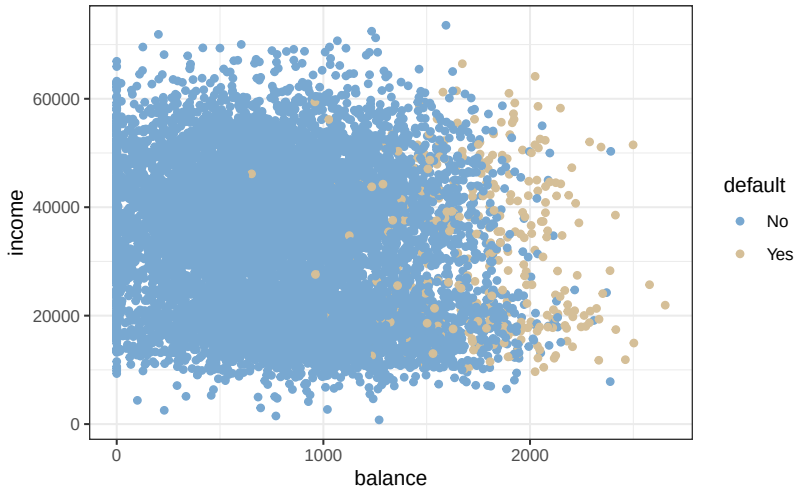
Classification

Classification

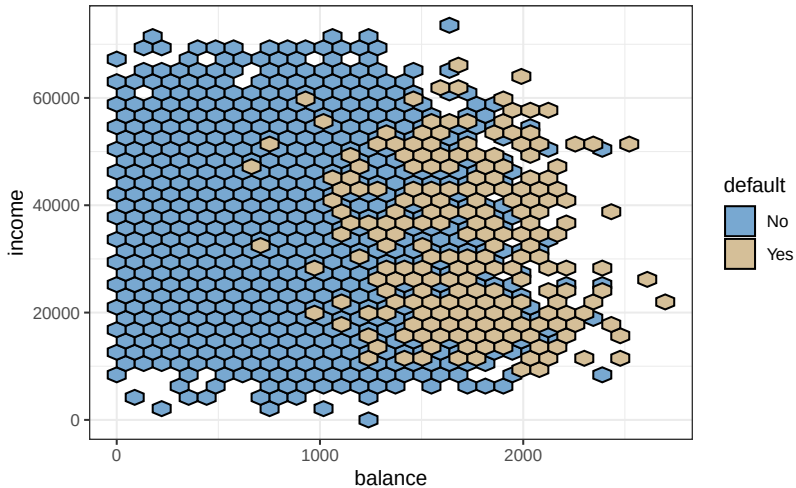
In classification, we are attempting to predict a categorical response variable.

Logistic Regression

Example



Example



General Linear Model

The “general” form of a general linear model consists of three parts:

1. A distribution of the observations,
2. a linear model, and
3. a link function that combines the two.

Logistic Regression

- **Distribution**

$$Y_i \sim \text{Bin}(n_i, \pi_i)$$

- **Linear model**

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}$$

- **Link function:**

$$\log \left(\frac{\pi_i}{1 - \pi_i} \right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}$$

Probabilities For Each Class

$$P(Y_i = 1|\mathbf{x}_i) = \frac{\exp(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip})}{1 + \exp(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip})}$$

$$P(Y_i = 0|\mathbf{x}_i) = \frac{1}{1 + \exp(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip})}$$

Estimating the Parameters

Likelihood

$$\ell(\boldsymbol{\beta}; \mathbf{y}) = \sum_{i=1}^m \{y_i \eta_i - n_i \log(1 + e^{\eta_i})\} + \text{Const},$$

where

$$\eta_i = \log \left(\frac{\pi_i}{1 - \pi_i} \right).$$

Score

$$S(\beta) = \left[\frac{\partial \ell}{\partial \beta_j} \right] = X^T (\mathbf{y} - \boldsymbol{\mu})$$

Fisher Information

$$\mathcal{I}(\beta) = X^T D X$$

where

$$D = \text{diag}(n_1\pi_1(1 - \pi_1), n_2\pi_2(1 - \pi_2), \dots, n_m\pi_m(1 - \pi_m)).$$

Newton-Raphson Algorithm

$$\beta^{(t+1)} = \beta^{(t)} - \left[S'(\beta^{(t)}) \right]^{-1} S(\beta^{(t)})$$

where

$$S'(\beta)$$

is the $p \times p$ matrix

$$\left[\frac{\partial S_j}{\partial \beta_k} \right] = \left[\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} \right].$$

Fisher Scoring

$$\beta^{(t+1)} = \beta^{(t)} + [\mathcal{I}(\beta^{(t)})]^{-1} \mathcal{S}(\beta^{(t)}).$$

For the logistic regression model, this can be simplified to

$$\beta^{(t+1)} = \beta^{(t)} + (X^T D^{(t)} X)^{-1} X^T (\mathbf{y} - \boldsymbol{\mu}^{(t)}).$$

Starting Point

$$\beta_0 = (X^T X)^{-1} X^T \mathbf{v}$$

where

$$v_i = \log \frac{y_i + 0.5}{n_i - y_i + 0.5}.$$

Beetles

```
## # A tibble: 8 x 3
##   Concentration Exposed   Dead
##           <dbl>   <dbl> <dbl>
## 1           1.69     59      6
## 2           1.72     60     13
## 3           1.76     62     18
## 4           1.78     62     28
## 5           1.81     63     52
## 6           1.84     59     53
## 7           1.86     62     61
## 8           1.88     60     60
```

Initial Point

```
get_b0 <- function(y, n, X){  
  v <- log((y + 0.5) / (n - y + 0.5))  
  b0 <- solve(t(X) %*% X) %*% t(X) %*% v  
  return(b0)  
}  
X <- cbind(1, beetles$Concentration)  
b0 <- get_b0(y = beetles$Dead, n = beetles$Exposed, X = X)  
b0
```

```
##           [,1]  
## [1,] -63.49803  
## [2,]  35.94502
```

Updating Step

```
update_b <- function(bt, y, n, X){  
  eta <- X %*% bt  
  pi <- exp(eta) / (1 + exp(eta))  
  mu <- n * pi  
  D <- diag(as.numeric(n * pi * (1 - pi)))  
  b <- bt + solve(t(X) %*% D %*% X) %*% t(X) %*% (y - mu)  
  return(b)  
}  
update_b(bt = b0, y = beetles$Dead, n = beetles$Exposed, X = X)  
  
##           [,1]  
## [1,] -59.56554  
## [2,]  33.57987
```

Iterate

```
## -59.56554 33.57987
```

```
## -60.53586 34.13457
```

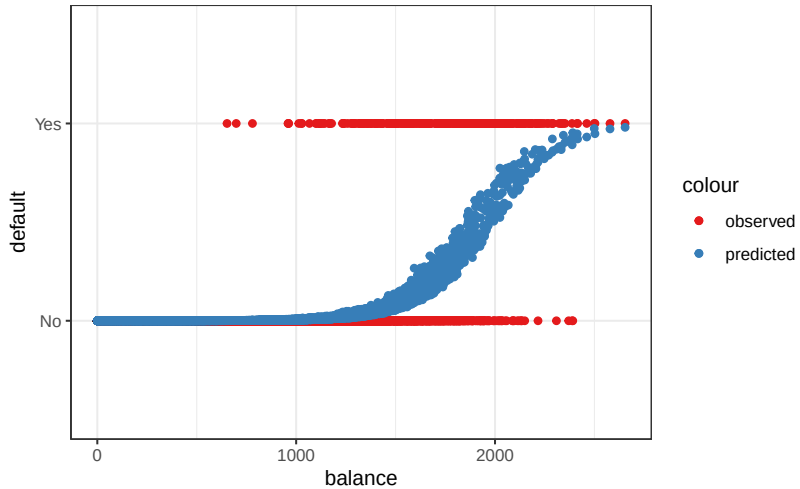
```
## -60.55595 34.146
```

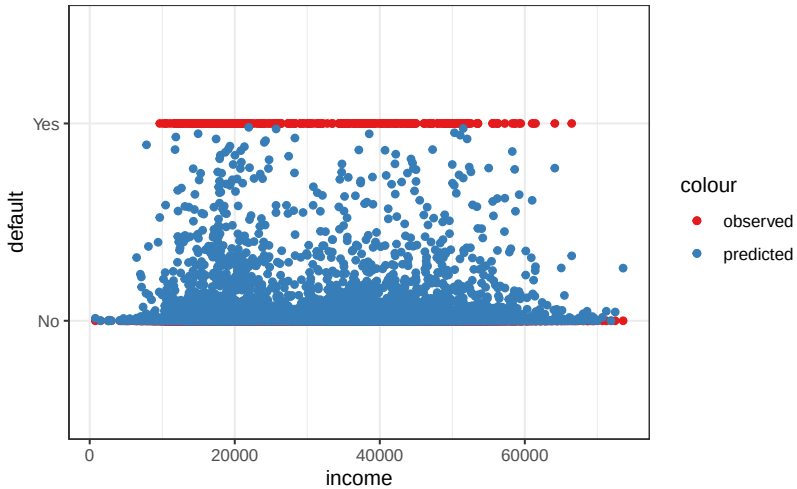
R Version

```
##
## Call:
## glm(formula = cbind(Dead, Exposed - Dead) ~ Concentration, family = binomial,
##      data = beetles)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.206  -0.226   1.012   1.359   1.678
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -60.556      5.151  -11.76  <2e-16 ***
## Concentration   34.146      2.893   11.80  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 287.499  on 7  degrees of freedom
## Residual deviance:  14.514  on 6  degrees of freedom
## AIC: 44.804
##
## Number of Fisher Scoring iterations: 4
```

Example (Loans)

```
##
## Call:
## stats::glm(formula = default ~ balance + income, family = stats::binomial,
##   data = data)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4725  -0.1444  -0.0574  -0.0211   3.7245
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.154e+01  4.348e-01 -26.545 < 2e-16 ***
## balance      5.647e-03  2.274e-04  24.836 < 2e-16 ***
## income       2.081e-05  4.985e-06   4.174 2.99e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2920.6  on 9999  degrees of freedom
## Residual deviance: 1579.0  on 9997  degrees of freedom
## AIC: 1585
##
## Number of Fisher Scoring iterations: 8
```





More Than Two Response Classes

Consider the case where we have J possible classes. We will consider the J th class as the baseline class. We then have

$$\log \frac{P(Y_i = j | \mathbf{x}_i)}{P(Y_i = J | \mathbf{x}_i)} = \beta_{j0} + \beta_{j1}x_{i1} + \dots + \beta_{jp}x_{ip}, j = 1, \dots, J.$$

Probability of Each Class

Rearranging gives

$$P(Y_i = j | \mathbf{x}_i) = \frac{\exp(\beta_{j0} + \beta_{j1}x_{i1} + \dots + \beta_{jp}x_{ip})}{1 + \sum_{k=1}^{J-1} \exp(\beta_{k0} + \beta_{k1}x_{i1} + \dots + \beta_{kp}x_{ip})},$$
$$j = 1, \dots, J-1$$

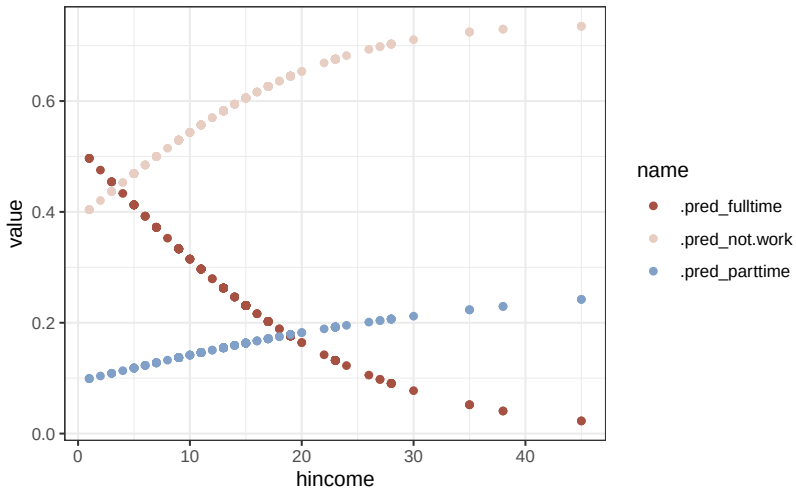
$$P(Y_i = J | \mathbf{x}_i) = \frac{1}{1 + \sum_{k=1}^{J-1} \exp(\beta_{k0} + \beta_{k1}x_{i1} + \dots + \beta_{kp}x_{ip})}$$

Example

##	partic	hincome	children	region
## 1	not.work	15	present	Ontario
## 2	not.work	13	present	Ontario
## 3	not.work	45	present	Ontario
## 4	not.work	23	present	Ontario
## 5	not.work	19	present	Ontario
## 6	not.work	7	present	Ontario

```
## parsnip model object
##
## Call:
## nnet::multinom(formula = partic ~ hincome, data = data, trace = FALSE)
##
## Coefficients:
##           (Intercept)      hincome
## not.work   -0.2896829  0.08354868
## parttime  -1.7009489  0.09024092
##
## Residual Deviance: 486.4405
## AIC: 494.4405
```

Prediction



ROC Curves



Available online at www.sciencedirect.com



Pattern Recognition Letters 27 (2006) 861–874

Pattern Recognition
Letters

www.elsevier.com/locate/patrec

An introduction to ROC analysis

Tom Fawcett

Institute for the Study of Learning and Expertise, 2164 Staunton Court, Palo Alto, CA 94306, USA

Available online 19 December 2005

Confusion Matrices

Person	favourite_pet	predicted_pet
Lewis	cat	cat
Gary	cat	cat
Jono	rabbit	rabbit
Mel	dog	dog
Rachael	dog	rabbit
Ben	dog	rabbit
Nigel	cat	dog
Giang	cat	rabbit
Josh	rabbit	cat

predicted_pet	cat	dog	rabbit
cat	2	0	1
dog	1	1	0
rabbit	1	2	1

predicted_cat	cat	not cat
cat	2	1
not cat	2	4

General Form

Predicted	True condition	
	Condition positive	Condition negative
Condition positive	True positive (TP)	False positive (FP)
Condition negative	False negative (FN)	True negative (TN)

Sensitivity / Specificity

$$\text{sensitivity} = \frac{TP}{TP + FN}$$

$$\text{specificity} = \frac{TN}{TN + FP}$$

Predicted	True condition	
	Condition positive	Condition negative
Condition positive	True positive (TP)	False positive (FP)
Condition negative	False negative (FN)	True negative (TN)

False Positive Rate / True Positive Rate

$$\text{False positive rate (FPR)} = \frac{FP}{FP + TN}$$

$$\text{True positive rate (TPR)} = \frac{TP}{TP + FN}$$

Predicted	True condition	
	Condition positive	Condition negative
Condition positive	True positive (TP)	False positive (FP)
Condition negative	False negative (FN)	True negative (TN)

Precision / Recall

$$\text{precision (PPV)} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

Predicted	True condition	
	Condition positive	Condition negative
Condition positive	True positive (TP)	False positive (FP)
Condition negative	False negative (FN)	True negative (TN)

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Predicted	True condition	
	Condition positive	Condition negative
Condition positive	True positive (TP)	False positive (FP)
Condition negative	False negative (FN)	True negative (TN)

F1 Score

The F1 score is the harmonic mean of precision and sensitivity.

$$\begin{aligned} F_1 &= 2 \frac{PPV \times TPR}{PPV + TPR} \\ &= \frac{2TP}{2TP + FP + FN} \end{aligned}$$

Cutoff 0.25

```
##           Truth
## Prediction   No   Yes
##           No 9658 292
##           Yes   9   41

## # A tibble: 2 x 3
##   .metric .estimator .estimate
##   <chr>    <chr>         <dbl>
## 1 sens    binary          0.999
## 2 spec    binary          0.123
```

Cutoff 0.5

```
##           Truth
## Prediction   No   Yes
##           No  9629  225
##           Yes   38  108

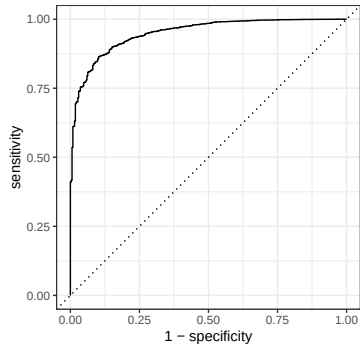
## # A tibble: 2 x 3
##   .metric .estimator .estimate
##   <chr>    <chr>         <dbl>
## 1 sens     binary         0.996
## 2 spec     binary         0.324
```

Cutoff 0.75

```
##           Truth
## Prediction   No   Yes
##           No  9468  148
##           Yes  199  185

## # A tibble: 2 x 3
##   .metric .estimator .estimate
##   <chr>    <chr>         <dbl>
## 1 sens     binary           0.979
## 2 spec     binary           0.556
```

ROC Curve



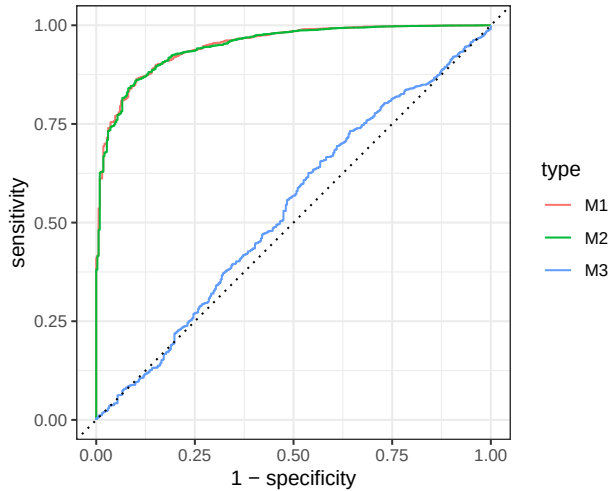
ROC Algorithm

```
1: function ROC(L, f, P, N)
2:    $L_{sorted} \leftarrow L$  sorted decreasing by  $f$  scores.
3:    $FP \leftarrow TP \leftarrow 0$ 
4:    $R \leftarrow \langle \rangle$ 
5:    $f_{prev} \leftarrow -\infty$ 
6:    $i \leftarrow 1$ 
7:   while  $i \leq |L_{sorted}|$  do
8:     if  $f(i) \neq f_{prev}$  then
9:       push ( $FP/N, TP/P$ ) onto  $R$ 
10:       $f_{prev} \leftarrow f(i)$ 
11:    end if
12:    if  $L_{sorted}[i]$  is a positive example then
13:       $TP \leftarrow TP + 1$ 
14:    else
15:       $FP \leftarrow FP + 1$ 
16:    end if
17:     $i \leftarrow i + 1$ 
18:  end while
19:  push ( $FP/N, TP/P$ ) onto  $R$ 
20:  return  $R$ 
21: end function
```

Comparing Models

```
M1 <- logistic_reg() %>%  
  set_engine("glm") %>%  
  fit(default ~ balance + income, data = Default)  
M2 <- logistic_reg() %>%  
  set_engine("glm") %>%  
  fit(default ~ balance, data = Default)  
M3 <- logistic_reg() %>%  
  set_engine("glm") %>%  
  fit(default ~ income, data = Default)
```

ROC Curves



AUC

```
1: function AUC(L, f, P, N)
2:    $L_{sorted} \leftarrow L$  sorted decreasing by  $f$  scores.
3:    $FP \leftarrow TP \leftarrow 0$ 
4:    $FP_{prev} \leftarrow TP_{prev} \leftarrow 0$ 
5:    $A \leftarrow 0$ 
6:    $f_{prev} \leftarrow -\infty$ 
7:    $i \leftarrow 1$ 
8:   while  $i \leq |L_{sorted}|$  do
9:     if  $f(i) \neq f_{prev}$  then
10:       $A \leftarrow A + \text{TRAPEZOID\_AREA}(FP, FP_{prev}, TP, TP_{prev})$ 
11:       $f_{prev} \leftarrow f(i)$ 
12:       $FP_{prev} \leftarrow FP$ 
13:       $TP_{prev} \leftarrow TP$ 
14:     end if
15:     if  $L_{sorted}[i]$  is a positive example then
16:        $TP \leftarrow TP + 1$ 
17:     else
18:        $FP \leftarrow FP + 1$ 
19:     end if
20:      $i \leftarrow i + 1$ 
21:   end while
22:    $A \leftarrow A + \text{TRAPEZOID\_AREA}(N, FP_{prev}, N, TP_{prev})$ 
23:    $A \leftarrow A / (P \times N)$ 
24:   return  $A$ 
25: end function
26: function TRAPEZOID_AREA(X1, X2, Y1, Y2)
27:   Base  $\leftarrow |X1 - X2|$ 
28:   Heightavg  $\leftarrow (Y1 + Y2) / 2$ 
29:   return Base  $\times$  Heightavg
30: end function
```


Example

```
## # A tibble: 3 x 4
##   type   .metric .estimator .estimate
##   <chr> <chr>    <chr>         <dbl>
## 1 M1     roc_auc  binary        0.949
## 2 M2     roc_auc  binary        0.948
## 3 M3     roc_auc  binary        0.533
```